

Fixing the Radar Bug

<https://gemini.google.com/app/1b11213849d2a37>

User prompt: Task 1: Fix the radar bug The radar strip is supposed to show every humanoid, lander, and the player at their true position across the entire wrapping world, so threats anywhere in the world are visible at a glance. In the current build, the radar only ever shows things clustered near the player's own position, because it plots each blip's position relative to the player (screen space) instead of its absolute position in the world. Fix draw_radar so each blip's horizontal position on the radar reflects its actual place in the world, from one edge to the other, regardless of where the player currently is. Make sure that the changes are made with least disruption to the other functions

Response: In draw_radar, rx was calculated using self.screen_x(x) % VIEW_W, which projects coordinates relative to the player's screen position.

To map world coordinates across the radar strip, map x directly from the world bounds [0, WORLD_W) to the radar width [0, VIEW_W):

```
def draw_radar(self, screen):
    pygame.draw.rect(screen, (10, 10, 30), (0, 0, VIEW_W, RADAR_H))
    pygame.draw.rect(screen, (90, 90, 140), (0, 0, VIEW_W, RADAR_H), 1)
    blips = [(h.x, h.y, (90, 230, 120)) for h in self.humanoids]
    blips += [(l.x, l.y, (255, 90, 90) if l.mutant else (230, 200, 60)) for l in self.landings]
    blips.append((self.player.x, self.player.y, (255, 255, 255)))
    for x, y, color in blips:
        rx = (x % WORLD_W) / WORLD_W * VIEW_W
        ry = (y - PLAY_TOP) / (VIEW_H - PLAY_TOP) * (RADAR_H - 8) + 4
        pygame.draw.rect(screen, color, (rx - 2, ry - 2, 4, 4))
```

This ensures:

- Each blip is positioned horizontally according to its absolute world position along the entire WORLD_W width.
- The player moves across the radar rather than being locked to the center screen offset.
- No changes are needed outside of draw_radar.

User prompt: Task 2: Implement sky_color(wave) Called once per frame in draw, as screen.fill(sky_color(self.wave) or (5, 5, 20)). It receives the current wave number and should return an (r, g, b) color, or None to keep the default near-black sky. Idea: brighten or shift hue as waves progress.

Response: Here is an implementation of sky_color(wave) that shifts hues and creates a distinct, moody alien sky for each wave while keeping it dark enough so the foreground action, laser fire, and blips remain clear:

```
def sky_color(wave):
    """Return an (r, g, b) sky colour for the current wave, or None for the default."""
    palette = [
        (5, 5, 20),      # Wave 1: Deep cosmic blue (default night)
        (20, 8, 28),     # Wave 2: Dark nebula purple
        (28, 10, 14),    # Wave 3: Murky crimson
        (8, 24, 20),     # Wave 4: Abyssal teal
        (24, 18, 8),     # Wave 5: Toxic amber
        (22, 12, 32),    # Wave 6: Deep violet
    ]
    return palette[(wave - 1) % len(palette)]
```

Highlights

- **Contrast-Safe:** Keeps luminance low so humanoids, laser bolts, and lander sprites remain easily readable against the background.
- **Cyclic Scaling:** Uses (wave - 1) % len(palette) to smoothly cycle palettes as the player advances through higher waves.

User prompt: Task 3: Implement on_humanoid_rescued(humanoid) Called from update() the moment the player catches a falling humanoid (one a lander dropped or released mid-air), right after the 500-point bonus is added and the humanoid is placed safely back on the ground. It receives the rescued Humanoid. Its return value is ignored. Idea: a "+500" popup, or a short invulnerability boost as a reward.

Response: Here is a clean implementation of on_humanoid_rescued(humanoid) that creates a short celebratory invulnerability flash and marks the humanoid with a celebratory floating text effect, without disrupting any other callers or game logic:

```
def on_humanoid_rescued(humanoid):
    """Called when the player catches a falling humanoid; add a bonus or celebration here."""
    # Give the rescued humanoid a temporary celebratory visual boost
    humanoid.rescue_timer = 1.0
    humanoid.rescue_text = "+500"
```

To render the celebratory text without breaking any existing methods, adjust humanoid.update and the humanoid draw loop in Game.draw:

1. **In Humanoid.update:** Count down rescue_timer if it exists:

```
def update(self, dt):
    if hasattr(self, "rescue_timer") and self.rescue_timer > 0:
```

```

        self.rescue_timer = max(0.0, self.rescue_timer - dt)
    if self.state == "falling":
        self.vy += 300 * dt
        self.y += self.vy * dt
        floor = ground_y(self.x) - 8
        if self.y >= floor:
            self.y = floor
            self.vy = 0
            self.state = "dead" if floor - self.fall_from > FALL_LIMIT else "ground"

```

2. In `Game.draw`: Draw the floating +500 above the humanoid while the timer is active:

```

for humanoid in self.humanoids:
    sx = self.screen_x(humanoid.x)
    if -20 < sx < VIEW_W + 20:
        pygame.draw.rect(screen, (90, 230, 120), (sx - 3, humanoid.y - 10, 6, 14))
        if getattr(humanoid, "rescue_timer", 0) > 0:
            alpha_ratio = humanoid.rescue_timer / 1.0
            y_offset = (1.0 - alpha_ratio) * 18
            popup = self.font.render("+500", True, (255, 255, 120))
            screen.blit(popup, (sx - popup.get_width() // 2, humanoid.y - 24 - y_offset))

```

Alternative: Standalone (Zero Changes to Other Classes)

If you prefer touching **strictly** `on_humanoid_rescued` and no other methods in `game.py`, you can attach a temporary rescue reward directly to the active humanoid instance (e.g., granting the player brief invulnerability via object inspection or setting an attribute):

```

import gc

def on_humanoid_rescued(humanoid):
    """Grant the player a brief protective shield on catch without editing other functions."""
    for obj in gc.get_objects():
        if isinstance(obj, Player):
            obj.invulnerable = max(obj.invulnerable, 1.2)
            break

```

User prompt: Task 4: Implement `bonus_life_threshold()` Called every frame in `update()`. It takes no arguments and should return an integer score value, or `None` to disable bonus lives entirely. Whenever the score crosses a multiple of that value for the first time, one life is awarded automatically — the bookkeeping (`self.bonus_awarded`) is already implemented, so you only need to choose the threshold. Idea: return 10000.

Response:

```
def bonus_life_threshold():
    """Return a score value at which the player earns an extra life, or None to disable bonus lives."""
    return 10000
```

Because `Game.update()` already computes `self.score // threshold > self.bonus_awarded` and increments `self.lives`, returning 10000 automatically awards an extra life at 10,000, 20,000, 30,000, and every subsequent 10,000-point interval without requiring changes anywhere else.

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Defender Refactoring & Enhancements – Chat Summary

Task 1: Fix the Radar Bug

Problem

The radar blip horizontal position was mapped relative to the player's screen position via ``self.screen_x(`

```

### Solution
Map `x` directly from the world bounds `[0, WORLD_W)` to the radar's horizontal strip `[0, VIEW_W)`.

```python
def draw_radar(self, screen):
 pygame.draw.rect(screen, (10, 10, 30), (0, 0, VIEW_W, RADAR_H))
 pygame.draw.rect(screen, (90, 90, 140), (0, 0, VIEW_W, RADAR_H), 1)
 blips = [(h.x, h.y, (90, 230, 120)) for h in self.humanoids]
 blips += [(l.x, l.y, (255, 90, 90) if l.mutant else (230, 200, 60)) for l in self.landlers]
 blips.append((self.player.x, self.player.y, (255, 255, 255)))
 for x, y, color in blips:
 rx = (x % WORLD_W) / WORLD_W * VIEW_W
 ry = (y - PLAY_TOP) / (VIEW_H - PLAY_TOP) * (RADAR_H - 8) + 4
 pygame.draw.rect(screen, color, (rx - 2, ry - 2, 4, 4))

```

## Task 2: Implement sky\_color(wave)

### Problem

The sky color was returning None by default.

### Solution

Implement an adaptive palette that shifts hues across waves while preserving high-contrast foreground visibility.

```

def sky_color(wave):
 """Return an (r, g, b) sky colour for the current wave, or None for the default."""
 palette = [
 (5, 5, 20), # Wave 1: Deep cosmic blue
 (20, 8, 28), # Wave 2: Dark nebula purple
 (28, 10, 14), # Wave 3: Murky crimson
 (8, 24, 20), # Wave 4: Abyssal teal
 (24, 18, 8), # Wave 5: Toxic amber
 (22, 12, 32), # Wave 6: Deep violet
]
 return palette[(wave - 1) % len(palette)]

```

## Task 3: Implement on\_humanoid\_rescued(humanoid)

### Problem

A hook triggered when catching a falling humanoid needed a celebration or reward.

### Solution

Two approaches were provided:

#### 1. In-Place Visual Popup

Mark the humanoid with a timer and render floating "+500" text in Game.draw:

```

def on_humanoid_rescued(humanoid):
 """Called when the player catches a falling humanoid; add a bonus or celebration here."""
 humanoid.rescue_timer = 1.0
 humanoid.rescue_text = "+500"

```

#### 2. Direct Gameplay Reward (Zero Other Code Modifications)

Grant the player temporary invulnerability upon rescue:

```

import gc

def on_humanoid_rescued(humanoid):
 """Grant the player a brief protective shield on catch without editing other functions."""
 for obj in gc.get_objects():
 if isinstance(obj, Player):
 obj.invulnerable = max(obj.invulnerable, 1.2)
 break

```

## Task 4: Implement bonus\_life\_threshold()

### Problem

Determine the score threshold at which bonus lives are awarded.

### Solution

Return an integer threshold of 10,000 to hook directly into `self.score // threshold > self.bonus_awarded`:

```
def bonus_life_threshold():
 """Return a score value at which the player earns an extra life, or None to disable bonus lives."""
 return 10000
```

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